Variable-amplitude Weyl calibration with a divergence source

2026-09-23

Statement and conventions

Let (M, g) be a smooth closed oriented four-manifold, and let ω be a smooth self-dual two-form with $|\omega|_g^2 = 2$. Let t be smooth with $0 < \tau \le |t| \le T$. Set

$$A = t\left(\frac{2}{3} I - \omega \otimes \omega\right), \qquad R = W_g^+ - A, \qquad E = \mathrm{Ric}_g - \sigma g, \qquad \sigma \in \{-1, 1\}. \tag{1}$$

Two-forms have the exterior-algebra norm, and A, R have the Hilbert–Schmidt endomorphism norm. The algebraic Weyl tensor associated with A is

$$\mathcal{W}(A)_{ijkl} = \sum_{a,b} A_{ab} (\epsilon_a)_{ij} (\epsilon_b)_{kl}, \quad J_{jkl} = \nabla^i \mathcal{W}(A)_{ijkl}, \quad |J|^2 = \sum_{j,k,l} J_{jkl}^2, \tag{2}$$

where (ϵ_a) is an orthonormal basis of Λ_g^+. In particular $|\mathcal{W}(A)|^2 = 4|A|^2$. Put

$$v = \tfrac{1}{3} \log |t|, \quad \widehat{g} = e^{2v} g, \quad \widehat{\omega} = e^{2v} \omega, \quad Q_X = \nabla_X \omega - dv \wedge \iota_X \omega + X^\flat \wedge \iota_{\nabla v} \omega. \tag{3}$$

Here $|Q|^2$ uses the exterior norm in the two-form slots. The connection transformation formula gives $Q = e^{-2v} \widehat{\nabla} \widehat{\omega}$. Define

$$d = \fint_M (|E|^2 + |R|^2), \qquad H = \fint_M |\nabla^2 A|^2, \qquad \phi = \sqrt{|t|}\, \omega, \qquad B_\phi = \fint_M |\Delta_H \phi|^2. \tag{4}$$

All unmarked averages and norms refer to g. The Hodge Laplacian is nonnegative. The derivatives in H are covariant derivatives on $\mathrm{End}(\Lambda_g^+)$.

Theorem 1 (exact identity and weak estimate). The following pointwise and integral identities hold:

$$\boxed{|J|^2 = 2t^2 |Q|^2 + 4t^2 |d\omega + \tfrac{2}{3} d \log |t| \wedge \omega|^2,} \tag{5}$$

$$\boxed{\int_M \left(|\widehat{\nabla} \widehat{\omega}|_{\widehat{g}}^2 + 2 |d\widehat{\omega}|_{\widehat{g}}^2 \right) dV_{\widehat{g}} = \frac{1}{2} \int_M |t|^{-4/3} |J|^2 dV_g.} \tag{6}$$

There are constants depending only on the displayed parameters such that

$$\fint_M |J|^2 \le C \sqrt{dH}, \qquad H \le C_T (1 + B_\phi + d), \tag{7}$$

and consequently

$$\boxed{\fint_M \left(|Q|^2 + 2|d\omega + \tfrac{2}{3} d\log|t| \wedge \omega|^2\right) \le C_{\tau,T}\sqrt{d(1 + B_\phi + d)}.} \tag{8}$$

Equivalently, the conformal energy in (6), divided by $\mathrm{Vol}_{\widehat{g}}(M)$, is bounded by the right-hand side of (8), with another $C_{\tau,T}$. No derivative of Ricci curvature, bound on W^-, Sobolev constant, injectivity radius, diameter, or lower volume bound is assumed. Neither dt nor $d\omega$ is assumed small.

The source $\Delta_H(\sqrt{|t|}\omega)$ includes amplitude derivatives. They have been included in an explicit budget, not discarded. Another sufficient budget is

$$B_\omega + D_t = \fint_M \left(|\Delta_H \omega|^2 + |\nabla^2 t|^2 + |dt|^4\right) : \qquad H \le C_T(1 + B_\omega + D_t + d). \tag{9}$$

The exact zero-source implication, $J = 0 \Rightarrow \widehat{\nabla}\widehat{\omega} = 0$, is the classical conformal Kähler mechanism. The quantitative result here is the explicit sourced identity and the weak near-Einstein estimate with its stated differential budget. An L^2 bound on a connection defect does not provide coordinates, a nearby Kähler metric, or geometric compactness.

Proof of the exact identity

The Levi-Civita transformation formula on a two-form is

$$e^{-2v}\widehat{\nabla}_X(e^{2v}\omega) = \nabla_X\omega - dv \wedge \iota_X\omega + X^\flat \wedge \iota_{\nabla v}\omega. \tag{10}$$

Thus $Q_X \in \Lambda^+_g$ and $\langle Q_X, \omega\rangle = 0$. Alternating (10) gives

$$\mathrm{Alt}\, Q = d\omega + 2dv \wedge \omega = e^{-2v} d\widehat{\omega}. \tag{11}$$

We calculate at a point in an oriented orthonormal coframe indexed $0, 1, 2, 3$. Let

$$\Omega_1 = e^{01} + e^{23}, \quad \Omega_2 = e^{02} + e^{31}, \quad \Omega_3 = e^{03} + e^{12}, \quad \omega = \Omega_1.$$

The frame of Λ^+ used for components is $\Omega_a/\sqrt{2}$. Write $\nabla_i\omega = a_i\Omega_2 + b_i\Omega_3$ and $h_i = \nabla_i t$. At the chosen point,

$$\nabla_i A = h_i\,\mathrm{diag}(-4/3, 2/3, 2/3) - 2t\begin{pmatrix} 0 & a_i & b_i \\ a_i & 0 & 0 \\ b_i & 0 & 0 \end{pmatrix}. \tag{12}$$

The coefficients $\widetilde{a}_i, \widetilde{b}_i$ of Q_i are the rows of

$$\begin{pmatrix} a_0 - h_3/(3t) & b_0 + h_2/(3t) \\ a_1 - h_2/(3t) & b_1 - h_3/(3t) \\ a_2 + h_1/(3t) & b_2 - h_0/(3t) \\ a_3 + h_0/(3t) & b_3 + h_1/(3t) \end{pmatrix}. \tag{13}$$

Substitute (12) in (2). Direct contraction of the three displayed Ω_a shows that the resulting J equals the divergence obtained from the purely angular array

$$-2t\begin{pmatrix}0 & \widetilde{a}_i & \widetilde{b}_i\\ \widetilde{a}_i & 0 & 0\\ \widetilde{b}_i & 0 & 0\end{pmatrix}. \tag{14}$$

This is an equality of tensors, not only their norms. In particular, for arbitrary column vectors a, b, let $V_1 = -2t(\Omega_2^T a + \Omega_3^T b)$, $V_2 = -2t\Omega_1^T a$, and $V_3 = -2t\Omega_1^T b$. The tensor represented by (14) is $J_{jkl} = \frac{1}{2}\sum_c (V_c)_j (\Omega_c)_{kl}$ with $a = \widetilde{a}, b = \widetilde{b}$. Since the Ω_c are orthogonal and have full component norm squared four,

$$|J|^2 = 4t^2\left(|a|^2 + |b|^2 + |\Omega_2^T a + \Omega_3^T b|^2\right). \tag{15}$$

Here $|Q|^2 = 2(|a|^2 + |b|^2)$ and $|\operatorname{Alt} Q|^2 = |\Omega_2^T a + \Omega_3^T b|^2$, proving (5). All twelve first-jet entries in (12) are arbitrary; no differential equation was used.

The conformal norm and volume rules give

$$|\widehat{\nabla}\widehat{\omega}|^2_{\widehat{g}} = e^{-2v}|Q|^2_g, \quad |d\widehat{\omega}|^2_{\widehat{g}} = e^{-2v}|\operatorname{Alt} Q|^2_g, \quad dV_{\widehat{g}} = e^{4v} dV_g.$$

Multiplying (5) by $|t|^{-4/3}/2 = e^{-4v}/2$ proves (6).

For a prescribed divergence source $S = \operatorname{div}\mathcal{W}(W^+)$, the exact tensor relation is $J = S - \operatorname{div}\mathcal{W}(R)$. Equation (5) therefore gives a direct strong-source estimate whenever this difference is controlled in L^2. The next argument obtains that control weakly from zeroth-order Ricci and Weyl errors and a second-order budget for A.

Weak Bianchi and the second-order budget

The contracted Bianchi identity and the parallel self-dual projection express $\operatorname{div}\mathcal{W}(W^+)$ as a fixed algebraic contraction of ∇E and $\nabla \operatorname{tr} E$. Since $\mathcal{W}(A) = \mathcal{W}(W^+) - \mathcal{W}(R)$, integration by parts gives, for every smooth three-tensor Z,

$$\left|\int_M \langle J, Z\rangle\right| \le C(\|E\|_2 + \|R\|_2)\|\nabla Z\|_2. \tag{16}$$

With $Z = J$, the parallel nature of (2) gives $|\nabla J| \le C|\nabla^2 A|$, proving the first inequality of (7). It does not assert an estimate for ∇E.

For completeness, the bounded-section estimates used below are

$$\int |\nabla z|^4 \le 16\|z\|_\infty^2 \int |\nabla^2 z|^2, \tag{17}$$

$$\int |\nabla^2 z|^2 \le 2\int |\nabla^* \nabla z|^2 + 32\|z\|_\infty^2 \int |\operatorname{Ric}|^2 + 33\|z\|_\infty^2 \int |F|^2. \tag{18}$$

Here F is the curvature of the metric bundle of z, with full two-index component norm. These estimates follow from integrating $\operatorname{div}(|\nabla z|^2\langle z, \nabla z\rangle)$ and the bundle Hessian identity. In the latter identity one first swaps the two Hessian slots and then commutes derivatives on ∇z; this produces Ricci curvature, $F\nabla z$, and $|Fz|^2/2$, with no derivative of F. Cauchy–Schwarz and Young's inequality, using (17), give (18). The full derivation appears in `round21_analytic.md`, Section 2; these inequalities are homogeneous in the volume and have no geometric compactness constant.

Apply them to $z = \phi = \sqrt{|t|}\omega$, whose pointwise norm is at most $\sqrt{2T}$. Since t has constant sign on each connected component,

$$A = \operatorname{sgn}(t)\left(\frac{|\phi|^2}{3} I - \phi \otimes \phi\right), \quad |A|^2 = \frac{8}{3} t^2. \tag{19}$$

The curvature of Λ^+ satisfies

$$|F^{\Lambda^+}|^2 \le C\left(|W^+|^2 + s^2 + |\operatorname{Ric}^\circ|^2\right). \tag{20}$$

Its W^- contribution vanishes, because the anti-self-dual Lie algebra factor acts trivially on Λ^+. Equations (1), (4), and (19) imply

$$⨍\left(|\operatorname{Ric}|^2 + |F^{\Lambda^+}|^2\right) \le C_T(1+d).$$

The Weitzenböck equation, for arbitrary ϕ, reads

$$\nabla^*\nabla\phi = \Delta_H\phi + 2W^+\phi - \frac{s}{3}\phi.$$

It supplies $⨍ |\nabla^*\nabla\phi|^2 \le C_T(1 + B_\phi + d)$. Using (17)–(18) therefore gives

$$⨍\left(|\nabla^2\phi|^2 + |\nabla\phi|^4\right) \le C_T(1 + B_\phi + d). \tag{21}$$

Differentiate the quadratic expression (19) twice: $|\nabla^2 A| \le C(|\phi||\nabla^2\phi| + |\nabla\phi|^2)$. This proves the second inequality of (7). Equations (5) and $|t| \ge \tau$ now prove (8). For the conformal average, use (6), $|t|^{-4/3} \le \tau^{-4/3}$, and $\operatorname{Vol}_{\widehat{g}}(M) \ge \tau^{4/3}\operatorname{Vol}_g(M)$.

Alternatively apply (17)–(18) to ω, whose norm is fixed, and differentiate $A = t(2I/3 - \omega\otimes\omega)$. This gives

$$|\nabla^2 A|^2 \le C_T\left(|\nabla^2\omega|^2 + |\nabla\omega|^4 + |\nabla^2 t|^2 + |dt|^2|\nabla\omega|^2\right).$$

Young's inequality bounds the final term by $(|dt|^4 + |\nabla\omega|^4)/2$, proving (9).

Endpoint, zeros, and relation to the earlier calibration

If $J = 0$, (5) forces $Q = 0$, so $\widehat{\omega}$ is parallel for $\widehat{g}$. Its norm is $\sqrt{2}$, and its associated orthogonal almost complex structure is parallel, hence integrable. Thus $(M, \widehat{g}, \widehat{\omega})$ is Kähler. When $R = 0$ and $\operatorname{Ric} = \sigma g$, contracted Bianchi gives $J = 0$ automatically. The simple Weyl eigenvalue is $-4t/3$. At the exact endpoint the Kähler scalar curvature is consequently

$$\widehat{s} = -8\operatorname{sgn}(t)|t|^{1/3}. \tag{22}$$

Without an Einstein or scalar-curvature equation, (5) cannot bound dt by J. The linear map from the twelve variables (dt, a, b) in (12) to J has rank eight. Its four-dimensional kernel is obtained by setting the eight entries of (13) to zero, leaving all four components of dt free. Thus an estimate $|dt| \le C|J|$ is already false at the first-jet level. A further equation is necessary to select a constant amplitude.

This theorem deliberately assumes both a globally supplied normalized form and a nonzero lower amplitude bound. If only the simple eigenline is supplied, local choices of ω differ by sign; the formula for A and all the energies descend. A global form can be chosen on the orientation double cover of that real line bundle. This cover addresses the sign of the eigensection, not zeros of t.

On an open nonzero-amplitude locus, (5) is valid locally, but no extension of $\widehat{g} = |t|^{2/3} g$ across $t = 0$ is asserted. The constants in (8) deteriorate as $\tau \downarrow 0$. No measure estimate for the zero set is hidden in the proof.

For the earlier model $A_{\psi,t} = t(|\psi|^2 I/3 - \psi \otimes \psi)$, normalization on a locus where $q = |\psi|^2 > 0$ gives $\omega = \sqrt{2/q}\,\psi$ and effective amplitude $\widetilde{t} = tq/2$. The theorem applies if that normalized data satisfies its hypotheses; the corresponding budget is the budget for $\sqrt{|\widetilde{t}|}\omega = \sqrt{|t|}\psi$. If ψ has zeros, this normalization does not produce a global application. The polynomial zero-set argument in the earlier fixed-parameter gap and the nonvanishing conformal theorem here address different regimes.

Primary-source comparison and verification

Derdziński, *Self-dual Kähler manifolds and Einstein manifolds of dimension four*, Compositio Mathematica 49 (1983), Proposition 5(i), pp. 420–421, proves the local conformal Kähler conclusion for harmonic W^+ with at most two eigenvalues, on $\{W^+ \neq 0\}$. His conformal metric is $|W^+|^{2/3} g$ up to a constant. Since $|A|^2 = 8t^2/3$, this is exactly the conformal factor in (3), up to a constant. Thus the zero-source conclusion is classical. Primary article.

LeBrun, *Einstein Metrics, Harmonic Forms, and Conformally Kähler Geometry*, arXiv:1903.00956v2, uses the conformal covariance of the harmonic-Weyl equation in Section 2 and obtains global alternatives in Theorem 2 under a near-symplectic harmonic-form hypothesis and a sign condition. Those hypotheses differ from the rank-one calibration and nonvanishing assumptions here. His treatment of zeros is not supplied by (5)–(8). Primary article.

The accompanying `line1_variable_calibration_check.py` checks the connection correction, all 64 components of (12)–(14), the norm identity on all twelve independent first-jet variables, the rank-eight symbol and its conformal kernel, and the conformal energy exponent. The integration by parts, curvature representation, and global consequences are proved above; symbolic checks do not establish these analytic assertions.

Local development baseline: commit `8113dd1`. The relevant user-supplied literature discussion is Section 6 of the theoretical-structure and literature-integration note in `line1_sources_20260923/`.